Brucellosis: A Comprehensive Overview

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Definition

Brucellosis (also called Malta fever, Mediterranean fever, or Undulant fever) is a zoonotic infection caused by *Brucella* species.

A systemic disease with acute or chronic presentations affecting multiple organs (e.g., liver, spleen, bones, heart, CNS).

Epidemiology

- **Global Burden:**
  - **Endemic** in Mediterranean, Middle East, Latin America, Africa, and parts of Asia.
  - **Incidence:** ~500,000 cases/year worldwide (underreported due to misdiagnosis).

Epidemiology

Transmission:

- **Primary sources:** Infected animals (cattle, goats, pigs, dogs).
- **Routes of infection:**
  - **Consumption** of raw milk/cheese.
  - **Direct contact** with infected animals (farmers, veterinarians).
  - **Inhalation** of aerosols (lab workers, slaughterhouses).
  - **Rarely:** Person-to-person (transplacental, breastfeeding, sexual).

Causative Organism

Bacteria:

- Gram-negative, facultative intracellular coccobacilli.
- Species:
  - *B. melitensis* (most common & virulent, goats/sheep).
  - *B. abortus* (cattle).
  - *B. suis* (pigs).
  - *B. canis* (dogs).

Causative Organism

Virulence Factors:	**LPS (Lipopolysaccharide)**: Weak endotoxin → evades immune detection.
	Type IV Secretion System (T4SS): Helps intracellular survival in macrophages.
	Superoxide dismutase & catalase: Resists oxidative killing.
	Stealth mechanisms: Avoids neutrophil & complement attack.

Pathogenesis

Entry: Bacteria enter via mucosa/skin → reach lymph nodes → bacteremia.

Dissemination: Spreads to **reticuloendothelial system** (liver, spleen, bone marrow).

Intracellular survival: Replicates in macrophages → granuloma formation.

Chronic infection: Persists due to immune evasion

Clinical Picture

Acute Brucellosis (1-2 months):	Fever (undulant, rising in afternoon), night sweats, fatigue.
	Arthralgia (sacroiliitis, spondylitis), myalgia, headache.
	Hepatosplenomegaly, lymphadenopathy.
Chronic Brucellosis (>1 year):	Recurrent fever, depression, chronic fatigue.
	Osteoarticular (vertebral osteomyelitis), neurological, cardiac complications.

Complications

- **Osteoarticular**: Spondylitis, sacroiliitis (most common).
- **Neurobrucellosis**: Meningitis, encephalitis, radiculopathy.
- **Endocarditis** (most fatal complication).
- **Hepatitis, orchitis/epididymitis, uveitis**.
- **Pregnancy**: Miscarriage, preterm birth, congenital brucellosis.

Diagnosis

Laboratory Tests:

- **Culture** (Gold standard but slow): Blood, bone marrow, CSF.
- **Serology**:
  - **Rose Bengal test** (screening).
  - **ELISA** (IgM = acute, IgG = chronic).
  - **Wright's agglutination test** (titer ≥1:160).
- **PCR**: Rapid & specific.

Imaging:

- X-ray (spondylitis), echocardiography (endocarditis).

Treatment

Combination therapy (to prevent relapse).

Duration: 6 weeks (uncomplicated), 3-6 months (neuro/endo-carditis).

First-line (Adults & Children >8 years):

- Doxycycline (100 mg BID) + Rifampin (600 mg OD) × 6 weeks.
- Alternative: Doxycycline + Streptomycin/Gentamicin (first 2 weeks).

Treatment

Children <8 years & Pregnant Women:

- Trimethoprim-sulfamethoxazole (TMP-SMX) + Rifampin.
- Avoid doxycycline (teeth discoloration) & aminoglycosides (renal risk).

Neurobrucellosis:

- Doxycycline + Rifampin + Ceftriaxone × 3-6 months.

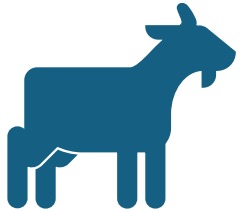
Endocarditis:

- Doxycycline + Rifampin + Aminoglycoside (surgery often needed).

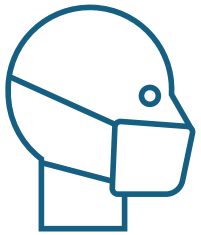
Prevention



Pasteurization of milk.



Vaccination of livestock (e.g., Rev-1 for goats).



Protective gear for high-risk workers

Summary



Brucellosis is a **zoonotic, multisystem disease** with chronic potential.



Diagnosis: Serology + PCR + Culture.



Treatment: Combination antibiotics (tailored to age/pregnancy/complications).



Prevention: Avoid unpasteurized dairy, vaccinate animals.

Thank you